

AI/ML Task 2: Model Comparison Report

Performance Evaluation & Feature Scaling Analysis

1. Objective

The objective of this task is to improve model performance using feature scaling, train multiple machine learning models, and compare their performance to select the optimal model for house price prediction.

2. Dataset

The **California Housing Dataset** from `scikit-learn` was utilized. It contains comprehensive demographic and geographic features including median income, house age, average room count, total population, and location coordinates. The target variable is median house value.

3. Preprocessing

Feature scaling was applied across all continuous predictor variables using `StandardScaler` to normalize data distribution to zero mean and unit variance. This prevents features with larger scales from dominating optimization algorithms and improves gradient convergence stability.

4. Models Used

- **Linear Regression:** Baseline parametric linear model.
- **Ridge Regression:** L2-regularized linear model to prevent overfitting.
- **Decision Tree Regressor:** Non-parametric tree-based model capable of capturing non-linear feature interactions.

5. Evaluation Metrics

Model performance was evaluated on the test set using standard regression metrics:

- **RMSE (Root Mean Squared Error):** Measures error magnitude in target units. Lower is better.
- **R² Score (Coefficient of Determination):** Quantifies variance explained by the model. Higher is better.

6. Performance Results

MODEL	RMSE	R ² SCORE
Linear Regression	0.7455	0.5757
Ridge Regression	0.7455	0.5758
Decision Tree Regressor	0.7242	0.5997

Best Performing Model: Decision Tree Regressor

The Decision Tree Regressor achieved the best overall metric scores, recording the lowest RMSE (0.7242) and the highest R² score (0.5997).

7. Conclusion

Feature scaling provided stable model inputs across baseline evaluations. The non-linear structure of the Decision Tree Regressor captured non-linear relationships present in the California Housing dataset more effectively than linear and regularized linear baselines, yielding stronger overall prediction accuracy.

8. Future Improvements

- Evaluate ensemble methods such as Random Forest, Gradient Boosting, and XGBoost.
- Perform systematic hyperparameter tuning using `GridSearchCV` or `RandomizedSearchCV`.
- Explore advanced feature engineering (e.g., interaction terms, spatial clustering).